Important Graph Mining Algorithms - Exam Summary

1. Louvain Algorithm

Main Idea: Used for community detection by maximizing modularity.

Important Formula
$Q = (1 / 2m) \Sigma_{ij} [A_{ij} - (k_i * k_j / 2m)] \delta(c_i, c_j)$

Focus on:

- Community detection
- Modularity
- Moving nodes between communities
- Building compressed graph

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2. Frequent Subgraph Mining

Main Idea: Finds graph patterns that appear frequently in a graph database.

Important Formula
Support(G') = Number of graphs containing subgraph G'

Focus on:

- Subgraph matching
- Support calculation
- Frequent patterns

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3. gSpan

Main Idea: Discovers frequent subgraphs using DFS codes without candidate generation.

Important Formula
Minimum DFS Code representation

Focus on:

- DFS Code
- Lexicographic order

- Pattern growth
- No candidate generation

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4. Apriori-based Graph Mining

Main Idea: Uses candidate generation and pruning similar to Apriori.

Important Formula
If a subgraph is infrequent, all its supergraphs are infrequent.

Focus on:

- Candidate generation
- Pruning
- Support threshold

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5. PageRank

Main Idea: Ranks nodes based on link importance.

Important Formula
$PR(A) = (1-d) + d \Sigma [PR(Ti) / C(Ti)]$

Focus on:

- Node ranking
- Incoming links
- Importance propagation

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6. Centrality Measures

Main Idea: Measures node importance in a graph.

Important Formula
Degree(v) = Number of connected edges

Focus on:

- Degree Centrality
- Betweenness Centrality

- Closeness Centrality
